

Pronouncing Acronyms in Esperanto

Marc van Oostendorp

Abstract

This chapter investigates how fluent Esperanto speakers pronounce acronyms, replicating and extending Torres-Tamarit and Martínez-Paricio's (2024) study of Spanish acronyms. Although Esperanto was designed to be maximally regular, no prescriptive rule was ever laid down for the pronunciation of letter sequences, making acronyms a revealing test case for emergent phonological constraints. We gathered data from 48 experienced speakers on their preferred pronunciations of acronyms varying in length and phonological structure. The results show a highly systematic pattern: longer acronyms that can form multi-syllabic words are almost always pronounced as whole words with regular penultimate stress, while two-letter acronyms are predominantly spelled out. Three-letter acronyms occupy a transitional zone, with monosyllabic options showing substantial variation. These patterns closely parallel the Spanish findings, suggesting that the same universal phonological constraints (including a prosodic minimality requirement and a preference for trochaic feet) govern acronym pronunciation regardless of whether the language is 'natural' or 'constructed'. The findings demonstrate that Esperanto speakers have developed unwritten conventions that mirror those of natural languages, underscoring the role of universal grammar in shaping even a planned language's phonology.

1. Introduction

Acronyms – abbreviations formed from the initial letters or groups of letters of words – present an interesting challenge for phonological theory and language users alike. Unlike ordinary words, acronyms are often novel strings of letters that speakers must pronounce without prior exposure or clear analogies in the lexicon. Different languages exhibit different strategies for pronouncing acronyms: some are read as words (e.g. NASA in English), while others are spelled out letter-by-letter (e.g. FBI).

The prosodic treatment of acronyms raises questions about how general phonological rules and constraints extend to such artificial inputs. Recently, Torres-Tamarit and Martínez-Paricio (2024) conducted a detailed study on Spanish acronyms, finding systematic patterns in how Spanish speakers assign stress and determine whether to pronounce an acronym as a word or as a sequence of letters. They discovered that Spanish acronym pronunciation obeys regular Spanish stress rules and is subject to a minimality constraint requiring a certain number of segments for an acronym to be realized as a prosodic word.

In this chapter, we report on a study that replicates and extends the Spanish findings in the context of Esperanto. Esperanto was designed to be maximally regular and easy to learn. Its grammar has very few exceptions; for example, word stress is fixed on the penultimate syllable of the word, and word formation is highly regular. However, Zamenhof did not explicitly account for every possible linguistic phenomenon.

Acronyms are one such area where no prescriptive rule was ever laid down: there is no official guideline on how to pronounce sequences of capital letters, and major Esperanto grammars offer little to no commentary on the matter. Nevertheless, Esperanto speakers do encounter and use acronyms (for instance, UEA for *Universala Esperanto-Asocio*, the Universal Esperanto Association), and in practice they have developed intuitions about their pronunciation. This situation provides a unique opportunity to observe “invisible regularization” in a language – that is, how a speech community spontaneously develops a coherent pattern for a linguistic task in the absence of explicit rules.

Our study gathered data from 48 fluent Esperanto speakers on their preferred pronunciations of various acronyms. By analyzing their choices, we aim to identify the emergent constraints governing Esperanto acronym pronunciation and compare them to the Spanish patterns described by Torres-Tamarit and Martínez-Paricio (2024). We show that Esperanto speakers exhibit strikingly similar behavior to Spanish speakers, despite the lack of any codified rule or shared native language influence.

The remainder of this chapter is organized as follows. In section 2, we review prior work on Spanish acronyms alongside relevant information about Esperanto phonology. The Methodology section describes our experiment with Esperanto speakers. In the Results, we detail the patterns observed in pronunciation mode (word vs. spelled-out) and stress placement, including quantitative summaries in tables. In the Discussion, we compare the Esperanto results to Spanish, interpret them within an OT framework as evidence for ad hoc faithfulness constraints, and consider the implications for the nature of Esperanto as a constructed language. Finally, we conclude by reflecting on the significance of these findings for interlinguistics and phonological theory.

2. Background

Recent work by Torres-Tamarit and Martínez-Paricio (2024) offers a formal account of how Spanish speakers pronounce acronyms, which serves as a point of comparison for our investigation in Esperanto. Their study combined a written survey and perception tests on Spanish acronyms, yielding two key generalizations. First, they found that Spanish acronyms follow the regular stress patterns of Spanish. In Spanish phonology, stress in polysyllabic words is largely predictable from the word’s ending: if a word ends in a vowel or *n/s*, the default stress is penultimate; otherwise (word ending in other consonants) the default stress is final.

Torres-Tamarit & Martínez-Paricio observed that when Spanish speakers read aloud an acronym that can be pronounced as a normal-sounding word, they assign stress as if it were an ordinary word following these rules. For example, an invented acronym like *INE* (*Instituto Nacional de Estadística* ends in a vowel, so it would naturally receive penultimate stress [ˈi.ne]. Second, they discovered that Spanish acronyms are subject to a prosodic size constraint: there appears to be an upper limit of three syllables for an

acronym pronounced as a single word, and importantly, a lower limit on size as well. Acronyms with four or more syllables tend to be avoided or simplified.

More interestingly for our purposes, acronyms comprising only one or two letters are generally not treated as standalone words in Spanish. Instead, speakers spell them out. In other words, there is a minimality requirement for acronyms that “must be expressed not in terms of syllable weight, but rather in terms of the number of segments”. Torres-Tamarit & Martínez-Paricio proposing a constraint within Optimality Theory (Prince and Smolensky 1993/2005) that effectively requires an acronym-word to contain at least three phonemic segments (consonants or vowels). An acronym like TV (two letters) in Spanish is pronounced *te uve* (naming each letter) rather than as a monosyllable [tev] or [TV], and an acronym like EU is pronounced “E-U” separately, not as [ew]. This behavior is not explained by ordinary Spanish phonotactics (since, for instance, [eu] could form a diphthong in Spanish in other contexts); instead, it points to a special constraint ensuring acronym words have a length.

Torres-Tamarit & Martínez-Paricio (2024)’s analysis thus introduced a well-formedness constraint specific to acronyms – requiring each letter of the acronym to correspond to its own syllable or at least prohibiting collapsing two letters into one syllable – to account for the reluctance to allow very short acronym-words. They note that such a constraint is unlikely to be a universal part of grammar (since it pertains to orthographic letters, a clearly learned and task-specific domain) and yet it appears to be consistently employed by Spanish speakers.

2.2 Esperanto Phonology and the Status of Acronyms

Esperanto’s prescriptive rules are simple and few in number. The most relevant rule proposed by Zamenhof for our study is the fixed stress rule: in Esperanto words of two or more syllables, stress always falls on the penultimate syllable (second-to-last).

Monosyllabic words (of which there are relatively few content words in Esperanto) are stressed on that single syllable by default. The phonotactic structure of Esperanto is also permissive enough to allow many sequences of consonants and vowels, with the main restriction being that every syllable must have a vowel (consonant clusters can occur, especially at syllable boundaries, but certain clusters might not be permissible as onsets, Van Oostendorp 1999).

These characteristics mean that many short sequences of letters can be pronounced as a phonologically legal Esperanto word. For instance, IFEK could be pronounced [ˈi.fek], TAR as [tar], KLA as [kla], etc., without violating any Esperanto phonotactic rule. On the other hand, a sequence like BFM has no vowel and thus cannot be pronounced as is in Esperanto; some strategy such as inserting vowel sounds or spelling out the letters by name would be required.

Crucially, Esperanto has no standard or widely taught convention for pronouncing acronyms. The comprehensive grammar PMEG (Plena Manlibro de Esperanta Gramatiko, Wennergren 2020) notes that abbreviations “can be pronounced by the names of the letters... Some abbreviations can be pronounced as words... various options are possible”, but it stops short of giving any firm rule, even warning that if one

were to pronounce an acronym as a single word, one should still apply the penultimate stress rule – yet “no one pronounces it that way.” In practice, experienced Esperanto speakers are familiar with certain common acronyms (usually organizational names like UEA, TEJO, ILEI). These are sometimes handled by spelling them out: e.g., UEA is typically said as [u.e. 'a], pronouncing each letter U-E-A separately. Some acronyms that form pronounceable strings are occasionally said as words – for instance, SAT (Sennacieca Asocio Tutmonda, a global Esperanto association) is often pronounced [sat] as if it were a regular word, rather than S-A-T, and similarly for TEJO = [tejo]. But these conventions are few, and for novel acronyms that speakers have never encountered, there is essentially a blank slate.

Many of our participants commented that they had “never thought about the issue” of how one ought to pronounce such letter strings, yet when prompted, they discovered they had strong intuitions about it. This scenario – fluent speakers applying their intuition to entirely artificial stimuli – allows us to observe the formation of phonological regularities almost *ex nihilo*.

Esperanto is an ideal case to examine the tension between “the unbridled creativity” of language invention and “the bridle of the mind” that ultimately governs any usable language. Despite being engineered for simplicity, Esperanto’s real-world usage inevitably involves phenomena (like acronyms) that were not explicitly engineered, thereby testing the language’s capacity to function “as if natural.” Indeed, Esperanto’s success over 137+ years – with an estimated ~2 million speakers (Eberhard, Simons & Fennig, 2024) and even a small number of native speakers – suggests that it is processed by its users with the same cognitive faculties used for natural languages.

3. Methodology

3.1 Participants

Forty-eight (48) fluent Esperanto speakers participated in the study. These included 45 members of the Akademio de Esperanto – an authoritative body composed of highly proficient Esperanto speakers (writers, linguists, educators, etc., from around the world) – and three additional fluent speakers of Esperanto.¹ The Akademio members were specifically chosen because they represent some of the most experienced and linguistically aware Esperanto speakers, often considered “guardian” users of the language. The participants spanned a variety of native-language backgrounds (Germanic, Romance, Slavic, and others), though there was a bias toward European L1s (as is common in the Esperanto community). Importantly, none of the participants was a native Spanish speaker, ensuring that any similarity to Spanish patterns cannot be attributed to transfer from a Spanish phonological system. Two participants identified as native Esperanto speakers (having spoken Esperanto from early childhood in the home); the remainder had learned Esperanto as a second language (often to a near-native level of fluency). Ages of participants ranged roughly from young adult to senior (exact ages were not collected). Participation was voluntary and uncompensated, and

¹ The author is currently also a member of the Akademio.

informed consent was obtained (the task was presented as a fun linguistic questionnaire).

3.2 Materials

We constructed a written questionnaire (administered online) consisting of a series of acronyms for the participants to pronounce. The stimuli were chosen to cover a range of structural types, in order to test the factors hypothesized to influence pronunciation (length in letters, presence or absence of vowels, phonotactic pronounceability, etc.). There were a total of 25 acronym items. Some of these were real acronyms borrowed from contexts like Spanish organizations or international scientific abbreviations (e.g., TALGO, CEPESA, ADIF, which are known acronyms in Spanish, or UEA, a known Esperanto acronym), while others were invented acronyms with no real-world meaning (e.g., IFEK, BFM, KLA, etc., created to ensure certain letter combinations). None of the acronyms corresponded to actual common words in Esperanto. The list included:

Two-letter acronyms: e.g., PA, IL, PO, AL. These are minimal-length acronyms with one consonant and one vowel (or two vowels in one case) – theoretically pronounceable as a monosyllabic word (e.g. pa, il) but also easily spelled out (P-A, I-L).

Three-letter acronyms: This category had several subtypes:

- Those with one vowel (thus potentially forming a one-syllable word, often with a consonant cluster) – e.g., TAR, KLA, BAS, FRE, FRO, NAL.
- Those with two vowels (which would form two syllables if pronounced as a word) – e.g., AES, EIN, TEI, EUC, IUC.
- Those with no vowel (consonant-only strings, not pronounceable as-is) – e.g., BFM was intended to represent this type (three consonants).

Four-letter acronyms: e.g., IFEK, INEF, FIEK (similar pattern with two vowels yielding two syllables), ADIF, PUAM, LOGS (some with two syllables as words, some with one syllable plus cluster).

Five-letter and six-letter acronyms: e.g., TALGO, CEPESA, MUNPAL, LOGSA (six and five letters respectively). These all contained at least two vowels, making them two or three syllables if read as a word.

The acronyms were presented in capital letters (to emphasize their status as letter sequences). For each item, participants were asked to imagine seeing this acronym in an Esperanto text or context and to indicate how they believe it should be pronounced in Esperanto.

4. Procedure

The questionnaire was administered electronically (via a form), and participants could complete it at their own pace. For each acronym item, the participants were instructed to write out the pronunciation in a way that made clear both the segmental form (whether they were pronouncing it as one continuous word or spelling it out letter by letter) and the placement of stress. We provided an example with the acronym ILEI (a real Esperanto acronym for Internacia Ligo de Esperantistaj Instruistoj, “International League of Esperanto Teachers”), explaining that one could answer in any of the following forms: “i lo e i” (if spelling out the letters by name, separated by spaces or

hyphens), “iléi” (if pronouncing as a single word, adding an acute accent to the stressed vowel), or the full phrase (the latter was not our interest but given for completeness). For each response, we coded two main pieces of information: (1) Pronunciation mode – whether the participant pronounced the acronym as a single whole word or as a sequence of letter-names – and (2) Stress placement – which syllable received the primary stress. The results were tallied for each acronym item, yielding the proportion of participants who chose each pronunciation strategy. We then aggregated the results by type of acronym (by length and structure) to observe broader patterns.

5. Results

5.1. Differences in size

Overall, the participants’ intuitions about acronym pronunciation were highly systematic and in line with our expectations: Longer acronyms that could form multi-syllable words were almost always pronounced as words with penultimate stress, very short acronyms (especially two-letter ones) were usually spelled out letter-by-letter, and intermediate cases (three-letter acronyms forming one-syllable or two-syllable words) showed a split in preferences, with a majority leaning one way depending on the specific conditions. Table 1 provides a summary of pronunciation preferences for a representative set of acronyms of different types, and we elaborate on each pattern below.

Each acronym is listed with its letter count. “Whole-word” indicates the percentage of participants who pronounced the acronym as a single word (applying Esperanto’s penultimate stress if multi-syllabic). “Letter-by-letter” indicates the percentage who spelled out the acronym (pronouncing each letter name separately; stress typically on the final letter name). Percentages are rounded to the nearest whole number.

Table 1. Pronunciation preferences by acronym type (selected examples).

Acronym stress)	Letters	Whole-word (penultimate stress)	Letter-by-letter (final
IFEK	4	88% (e.g. [í.fek])	12% (e.g. i-fo-kó)
BFM	3	0% (n/a – no vowel)	100% (e.g. bo-fo-mó)
PA	2	31% (e.g. [pa])	69% (e.g. po a)
IL	2	27% (e.g. [il])	73% (e.g. i lo)
TAR	3	64% (e.g. [tar])	36% (e.g. to ró)
KLA	3	85% (e.g. [kla])	15% (e.g. ko-lá)
FRE	3	34% (e.g. [fre])	66% (e.g. fo-ré)

Examining the results in Table 1 and the broader dataset:

Multi-syllabic acronyms (four or more letters, or three letters with two vowels): These acronyms overwhelmingly tended to be pronounced as a single word following the normal Esperanto stress rule. For example, IFEK (a four-letter string with two vowels) was pronounced [í.fek] (two syllables, stress on “i”) by 88% of participants, adhering to penultimate stress. Only a small minority (12%) chose to spell “IFEK” out as i-fo-e-ko.

Similarly, five-letter acronyms like TALGO and CEPSE (each forming two syllables as a word) were almost unanimously treated as words: 95% of respondents pronounced “TALGO” as [tál.go] (with tál stressed) and “CEPSE” as [cép.sa] (with cép stressed). Hardly anyone spelled these out (only 5% for TALGO, 7% for CEPSE, respectively). Even an unfamiliar six-letter string like MUNPAL was handled with ease: 97% gave a pronunciation “múnpall” (stressing mun, the penultimate syllable) or equivalent. The consistent application of penultimate stress in these cases confirms that speakers extend Esperanto’s primary word-stress rule to acronyms that they treat as lexical items.

Acronyms impossible to pronounce as words (no inherent vowel): In line with expectations, when an acronym’s letters provided no viable vowel sequence, participants uniformly resorted to spelling it out. The clearest case is the hypothetical acronym BFM: of the 48 participants, 100% spelled “BFM” as letter name, “bo-fo-mó”. Here “bo.fo.mó” has an accent on “mó” indicating that the last letter name carries the main stress. This aligns with what one might expect from phrase-level accent: when spelling out a sequence of words/letters in Esperanto (as in many languages), the last element often has a focal stress or is spoken with a falling intonation. Notably, participants did not try to force a penultimate stress as if “bofomo” were one word – they intuitively treated the sequence as separate chunks and naturally stressed the final one (some wrote it as separate words “bo fo mó”, others hyphenated, but in both cases the last “mó” was marked as stressed).

Two-letter acronyms: The vast majority of participants prefer to spell out two-letter acronyms rather than pronounce them as a single syllable. As Table 1 shows, for an acronym like PA (P-A), 69% of respondents answered with “po a” (or “pó-o”, etc., indicating P = po, A = a) and only 31% responded with “pa” as a one-syllable word. For IL (I-L), 73% gave “i lo” and 27% gave “il”. We observed the same trend for other two-letter combinations: PO (P-O) was spelled out by ~69% (e.g. pó-o) and pronounced as [po] by ~31%; AL (A-L) was spelled out by ~75% (often as á-lo or a-ló) and pronounced [al] by ~25%. In the cases of “AL” and “PO”, which as written forms coincide with actual Esperanto function words (“al” means “to” as a preposition, “po” is a distributive preposition meaning “at the rate of”), some participants noted that pronouncing them as the existing words felt strange in an acronym context – perhaps a minor semantic interference – but even aside from that, the tendency was clearly toward spelling them. The two-letter acronym data, taken together, indicate a strong avoidance of treating a two-letter string as a standalone prosodic word in Esperanto. Even though phonetically nothing prevents [pa] or [il] or [al] from being uttered, speakers appear to instinctively find a two-letter acronym “too short” or insufficient as a word in itself. Instead, they opt to pronounce each letter separately, yielding a disyllabic phrase. Stress placement in spelled-out two-letter acronyms was somewhat variable: about two-thirds of those who spelled “AL” or “PA” out put the stress on the second letter (saying pó ó or á ló, effectively accenting the final letter name), while the remaining one-third stressed the first letter name (e.g. “pó o”, “á lo”).

Three-letter acronyms (monosyllabic): This category is particularly revealing, as it sits at the boundary of the minimality constraint that speakers seem to observe. For three-letter strings that contain exactly one vowel (hence form one syllable if pronounced together), we observed a split, though usually with a majority favoring the single-word pronunciation. For example, TAR (T-A-R) was pronounced as a one-syllable word [tar] by

Commented [xx1]: PO [?]

about 64% of participants, and spelled out (“to a ro”) by 36%. KLA (K-L-A) behaved even more like a word: approximately 85% said [kla] (which sounds like a plausible syllable in Esperanto, similar to the beginning of klar “clear”), and only 15% spelled it out. Participants who treated “KLA” as [kla] all stressed it inherently on that one syllable (some wrote “klá” with an accent to indicate the stress, although in a one-syllable word the concept of stress is trivial). Those who spelled “KLA” tended to produce two syllables (“ko-lá” or “ko-lo a”) because K = ko, L = lo, A = a – interestingly, some might combine the last two into “la” yielding “ko-lá” (stress on la), effectively making it two syllables, while others might say all three separately “ko lo a” which is three syllables; the data showed both representations, but they were few in number anyway. FRE (F-R-E) and FRO (F-R-O) showed more people spelling out than pronouncing as a word. For FRE, only ~34% said [fre], whereas 66% spelled it (“fo-ré-e” or “fo-ró-e”). For FRO, it was roughly an even split (47% [fro] vs. 53% “fo-ro-o”). It is not clear what might explain the difference between, say, TAR/KLA (mostly words) and FRE/FRO (many spelled-out forms).

The important point is that three-letter acronyms with one vowel generally had a substantial proportion of speakers (often a majority) willing to pronounce them as a single syllable word – unlike two-letter acronyms. This underscores the role of acronym length (in letters) as a deciding factor, exactly as found for Spanish. In effect, Esperanto speakers also treated “three letters” as the minimum for an acronym to be “word-like”. When asked informally, some participants reasoned that pronouncing two letters as one syllable might “hide” one of the letters or make it less audible, whereas with three consonants you still hear enough of the combination to recognize each letter’s contribution. This intuition aligns with the idea of a constraint ensuring each letter is salient: with two letters merged into one syllable, each letter’s independent identity might be perceived as undermined, but three letters can sometimes still be discerned (though this rationale is subtle and was not articulated by everyone).

Three-letter acronyms (multi-syllabic as words): For three-letter strings that contain two vowels (and hence would naturally be two syllables if read as a word), we saw a different pattern: these behaved more like the multi-syllabic acronyms in that a majority pronounced them as words, but the majority was not as overwhelming as with four-letter words like IFEK. For example, AES (A-E-S) can be pronounced as two syllables [a.‘es] (essentially “a-es”) or spelled out “a e so”. About 58% gave a two-syllable word pronunciation (often writing “áes” or “aés” – interestingly some ambiguity whether they treat it as hiatus or a diphthong; most treated A and E as separate syllables), and ~42% spelled it out (e.g. “a-és”). EIN (E-I-N) is a complex case because “ein” could be [ajn] if someone treated <ein> like in German or English, but in Esperanto orthography E and I would typically be separate syllables unless i were a semivowel. Participants were split: some wrote “éin (1 silabo)” treating ein as one syllable (perhaps analogizing to a diphthong [ejn]), while others wrote “e-in (2 silabo)”, treating it as two syllables. Counting all approaches, 59% treated EIN as more than one syllable (either spelled out fully as letters e-i-nó by some, or as two-syllable éin by others), and 41% tried to pronounce it as a one-syllable ein. This particular item showed confusion, but importantly, nearly everyone still followed the principle of making sure the stress was penultimate if they had two syllables (those who wrote é-in effectively stressed e, the first of two syllables, which is penultimate; those who spelled it out put stress on the

final letter name or penultimate depending on interpretation). TEI (T-E-I) similarly had a mix: many pronounced it as [ˈte.i] (two syllables “te-i”), some as a single syllable like [tej] (writing “téi” perhaps to indicate a diphthong), and some spelled “té-i”.

Despite the variability, one constant was observed: if an acronym could be pronounced as an Esperanto-compliant word without breaking obvious rules, a good proportion of speakers would do so. Spelling out was usually the option when pronunciation as a word was awkward either due to extreme brevity or uncertainty in vowel combinations.

To summarize the preference patterns: Acronyms with three or more letters were usually pronounced as whole words, unless pronunciation was impeded by missing vowels or difficult clusters, in which case spelling out was the fallback. Acronyms with only two letters were usually spelled out, reflecting a disfavor for monosyllabic acronym-words consisting of just two segments. These tendencies in Esperanto are remarkably similar to those documented for Spanish acronyms. Spanish speakers likewise generally pronounce longer acronyms as words (e.g., UNESCO as [uˈnesko]) and spell out very short ones (e.g., EU as E-U), positing a minimal size for acronym words. The Esperanto data reinforce that this is not an arbitrary convention of Spanish, but potentially a broader preference in how people treat letter-sequences phonologically.

5.2 Variability and unanimity

One of the notable findings is that judgments were never completely unanimous for any acronym item, yet they were far from random – in each case one option was chosen by a clear majority. The closest to unanimity we saw was in cases like TALGO or MUNPAL where 95%+ pronounced them as words; and BFM where 100% spelled out. In other cases (especially three-letter acronyms), the split might be, say, 2:1 or 3:2 in favor of one method. Participants themselves commented on this variability: several noted that they had been “torn” between two possible pronunciations for a given item and had to decide which felt more natural. This was expected – acronym pronunciation is not something fixed by rule or long convention in Esperanto, so a certain amount of free variation is natural. Nevertheless, the existence of very clear trends (e.g., nearly everyone leaning toward spelling for two-letter items, or leaning toward word pronunciation for longer items) indicates an emerging consensus or at least a shared bias. We did not find any pattern that those who, say, spelled out one item were consistently more likely to spell out all items; rather, most individuals’ responses varied item by item in line with the overall tendencies for that item. This suggests that speakers are sensitive to the structural properties of each acronym (length, presence of vowels, etc.) in a similar way. We also examined whether a speaker’s native language influenced their choices – for instance, perhaps a speaker whose L1 does not allow complex consonant clusters might be more inclined to spell out acronyms with such clusters. We did not find evidence for strong L1 influence; even speakers of languages like French (who might not be used to words ending in certain consonants) or languages with fixed initial stress, for example, generally conformed to the Esperanto-like pattern rather than imposing their L1 phonology. The patterns appeared independent of the speakers’ native languages. This aligns with the idea that in pronouncing Esperanto acronyms,

speakers were guided by either universal or Esperanto-specific considerations rather than by transferring rules from their mother tongues. Indeed, many participants have extensive experience using Esperanto and tend to “think” in Esperanto norms when speaking it, as evidenced by their consistent use of penultimate stress and Esperanto letter names.

6. Discussion

6.1. Comparison with Spanish and implications for phonology

One of the most striking outcomes of this study is how closely the Esperanto results parallel the Spanish results of Torres-Tamarit & Martínez-Paricio (2024). In essence, the same phonological generalizations (a, b, c below) hold in both languages:

(a) A preference for trochaic, multi-syllabic pronunciations: Both Esperanto and Spanish speakers tend to pronounce an acronym as a single prosodic word with a trochaic (penultimate) stress pattern whenever the letter string permits it. In Spanish, this meant adhering to the language’s stress rules (usually penultimate stress for acronym-words, since many end in a vowel or -s, -n). In Esperanto, it meant consistently applying penultimate stress to any acronym treated as a word (since that is the invariant rule in Esperanto). Our data showed this clearly – no matter how unfamiliar the acronym, if it was pronounceable as a two- or three-syllable word, speakers pronounced it with the stress on the penultimate syllable, just as they would an ordinary Esperanto word. This indicates that the default prosodic structure of the language (the “ordinary language” footing) is extended to acronyms.

(b) The presence of consistent but not unanimous choices, i.e., variation with a clear trend: In Spanish, Torres-Tamarit & Martínez Paricio (2024) note that not all speakers always agreed (there was variation), but typically one outcome was preferred by most, and crucially, the variation did not obscure the overall pattern. We observed the same in Esperanto. For each acronym, there was usually a dominant pronunciation mode (word vs. spelled) that won out, and a minority alternative. The minority choices, however, never defied the fundamental constraints at play – they were usually the “other obvious option” rather than something out of left field.

(c) Independence from native language influence: Torres-Tamarit & Martínez-Paricio’s Spanish study was done with native Spanish speakers, so L1 influence wasn’t a question there. Our Esperanto study involved speakers with various L1s, and we found that patterns did not differ systematically by L1. An English-speaker, a Russian-speaker, and a Japanese-speaker in our sample (to pick illustrative backgrounds) all tended to follow the same decision-making: e.g., all would likely spell out “IL” and pronounce “TALGO” as a word. This speaks to these acronym-handling strategies being somewhat universal or driven by the target language’s structure rather than one’s first language. It also supports Torres-Tamarit & Martínez Paricio (2024)’s implication that the phenomena we’re seeing are not accidents of a particular language’s history, but rather how human linguistic competence deals with acronyms given certain shared parameters (like stress patterns and the salience of segmental units).

6.2 Significance for our understanding of constructed languages

Beyond phonological theory, these results have implications for how we view Esperanto's linguistic development. Esperanto is often discussed in terms of naturalness vs. artificiality. Some scholars (and critics) have questioned whether an invented language can ever function with the full complexity and nuance of a natural language – whether its “extreme regularity” might actually be a weakness that limits expressiveness or whether speakers will inevitably introduce irregularities or patterns not foreseen by the creator. Our study provides a concrete example of Esperanto spontaneously evolving a convention (or incipient rule) in an area that was not explicitly designed: acronym pronunciation. We see that Esperanto speakers handled acronyms in a very natural way, i.e., they employed strategies that are cognitively sensible and even mirror those in a natural language (Spanish).

. This addresses the question “Can a language function if someone decided to do things differently on the drawing board?” – yes, it can, because the speaker community will supply whatever unwritten rules are needed to make it function. The “invisible regularization” of acronym pronunciation is evidence that Esperanto's user base collectively creates norms in the same way a natural language community does: through usage and analogy, guided by general linguistic instincts. Far from being a rigid system where only what's prescribed exists, Esperanto shows fluid adaptation. In essence, Esperanto behaves as a living organism would (linguistically speaking), growing new “organs” to handle new situations, albeit within the limits of general linguistic architecture. It's also notable that the constraint (or rule) that emerged – do not pronounce two-letter acronyms as a single syllable – actually adds a layer of complexity or irregularity that wasn't originally in Esperanto's simple rule set. Zamenhof didn't say “words under a certain length must be spelled out” – that's something the speakers have effectively added. This resonates with the idea put forth by Gertjan Postma (p.c.) that certain irregularities or complexities in natural languages might not be arbitrary, but necessary for functionality. In natural languages, for example, extremely short content words are often avoided or augmented (English has ‘a’, ‘I’ as exceptions, but many languages require at least CVC or CV structure for content words). Esperanto's core grammar did not impose a minimum word length beyond one syllable, but here we see a de facto enforcement of a two-syllable minimum in this specific context of acronyms. One could interpret this as the Esperanto community subconsciously introducing a functional irregularity to ensure clarity – much as natural languages have done over millennia.

From an interlinguistic perspective, our data show that even when faced with unnatural input (like random letter strings), Esperanto speakers keep applying that natural-like phonology and only innovate as needed. It is also worth noting that our participant group – members of the Akademio – are precisely those who might influence or codify usage norms for Esperanto in the future. The fact that they largely agree on these patterns means that if the question “How should we pronounce acronyms?” ever formally comes before the Akademio, they have a ready answer based on intuition: Pronounce them in the most Esperanto-like way possible (with penultimate stress) when you can, but if the acronym is very short or unpronounceable, say the letters individually. Indeed, one can find an entry in Bertilo Wennergren's PMEG (I used the 2020 edition) acknowledging multiple possibilities and essentially echoing what we found – that you can pronounce some as words and others you pronounce letter by

letter, and that it's normal that "various options are possible". Wennergren even specifically notes that if you were to combine letters into one word, you would stress penult (giving a hypothetical example) but emphasizes that in reality people don't do that for sequences that aren't obviously word-like. Our empirical study gives weight to those observations and refines them with the notion of a letter-count threshold.

7. Conclusion

This chapter presents research on how fluent Esperanto speakers pronounce acronyms, examining the phonological constraints involved and comparing them to Spanish. The study of 48 experienced Esperanto speakers revealed systematic pronunciation patterns. These patterns mirror those in Spanish, suggesting they stem from general phonological strategies rather than language-specific features. The researchers propose that Esperanto speakers employ a task-specific faithfulness constraint ensuring each letter remains aurally distinct, preventing very short acronyms from collapsing into a single syllable.

The research has broader implications, showing that constructed languages develop unwritten conventions similarly to natural languages. Esperanto's handling of acronyms demonstrates the language's natural evolution, with speakers instinctively imposing structures (like trochaic stress patterns) that align with cross-linguistic norms. This case exemplifies how users of a lingua franca navigate new linguistic territory by balancing established rules with innovative constraint-building to ensure effective communication.

References

- Benua, L. (1997). *Transderivational Identity: Phonological Relations between Words*. (Doctoral dissertation, University of Massachusetts Amherst).
- Eberhard, D. M., Simons, G. F., & Fennig, C. D. (2024). *Ethnologue: Languages of the World* (27th ed.). Dallas, TX: SIL International. (Entry on Esperanto)
- Itô, J., & Mester, A. (1995). Japanese phonology. In J. Goldsmith (Ed.), *The Handbook of Phonological Theory* (pp. 817–838). Blackwell.
- Malik-Moraleda, S., Taliaferro, M., Shannon, S., Fedorenko, E., & Gibson, E. (2025). Constructed languages are processed by the same brain mechanisms as natural languages. *Proceedings of the National Academy of Sciences*, 122(31), e2313473122. <https://doi.org/10.1073/pnas.2313473122>
- Prince, A., & Smolensky, P. (2004). *Optimality Theory: Constraint Interaction in Generative Grammar*. Malden, MA: Blackwell. (Original work published 1993)
- Torres-Tamarit, F., & Martínez-Paricio, V. (2024). The prosody of Spanish acronyms. *Natural Language & Linguistic Theory*, 42(2), 815–848. <https://doi.org/10.1007/s11049-023-09599-w>
- Van Oostendorp, M. (1999). Syllable Structure in Esperanto as an Instantiation of Universal Phonology. *Esperantologio/Esperanto Studies* 1:51-88.

Wennergren, B. (2020). Plena Manlibro de Esperanta Gramatiko (PMEG). Accessed from <https://bertilow.com/pmeg> (Section on abbreviations and acronyms).